

Environmental and Development Economics

Module 3 - How does the environment affect development?

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Lecture 4

How does the environment affect economic development?

Housekeeping

- ▶ Come talk to me about your research idea
 - ▶ Write half a page every couple days
- ▶ Start pinning down a paper to replicate
 - ▶ And think about extension (see example)
- ▶ Next class, you have TWO options for the presentation paper
- ▶ See VoxDevLit on air pollution!

Environment & Development Seminars

- ▶ **[New]** JEEP Seminar (12:30-2pm, Ruttan 119)
 - ▶ Oct 5th: Andy Hultgren (UIUC) presenting on ag. productivity and environment
 - ▶ Lunch served!

- ▶ APEC Seminar (Fridays, 10:30-12pm, Ruttan 119)
 - ▶ Sheetal Sekhri (UVA): Sept 25th, 2026
 - ▶ Molly Lipscomb (UVA): Feb 5th, 2027
 - ▶ Adam Storeygard (Tufts): Feb 26th, 2027
 - ▶ Maulik Jagnani (Tufts): March 19th, 2027

- ▶ **Guiding question:** how does the environment affect economic development?
- ▶ Start identifying envirodevonomics parameters
- ▶ Today's focus: health effects
- ▶ Next time: productivity and direct income effects

Recall our model of environment and development

$$MWTP_e = \frac{\lambda_e}{\lambda_y} = \frac{1}{\lambda_y} \left(\frac{\partial u}{\partial e} + \frac{\partial u}{\partial h} \frac{\partial h}{\partial e} \right) + \frac{\partial \Delta y}{\partial e} + \frac{\partial \Delta y}{\partial h} \frac{\partial h}{\partial e}$$

- ▶ Most research in environment/development economics seeks to identify $\frac{\partial h}{\partial e}$
 - ▶ At this point, $\frac{\partial h}{\partial e} > 0$ is unambiguous
 - ▶ Focus is on **magnitudes**
 - ▶ Less work on mechanisms and moderators
- ▶ What about indirect effects via productivity and health ($\frac{\partial \Delta y}{\partial h} \frac{\partial h}{\partial e}$)?
 - ▶ This is the research frontier
 - ▶ Next lecture

Existing Work to Date

- ▶ Largely from developed countries (see Janet Currie papers)
 - ▶ Review paper: “What Do We Know About Short- and Long-Term Effects of Early-Life Exposure to Pollution?” (Currie et al. 2014)
- ▶ Less in developing countries
 - ▶ Most benchmark magnitudes: Bharadwaj and Eberhard (2008); Jayachandran (2009); Chen et al. (2013); Greenstone and Hanna (2014); Arceo et al. (2016); Cesur et al. (2016); He et al. (2016); Barwick et al. (2018); Heft-Neal et al. (2018); Chang et al. (2019); Heft-Neal et al. (2019); Pullabhotla (2019); Bombardini and Li (2020); Fan et al. (2020); He et al. (2020); Adhvaryu et al. (Forthcoming)
- ▶ Less work on non-health effects
 - ▶ See Aguilar-Gomez et al. (2022) for review
 - ▶ Zarate-Barrera (2026): pollution and crime
- ▶ Empirical innovations: new quasi-experiments for identifying pollution impacts
 - ▶ Thermal inversion, wind direction IV (see Oliva & Deryugina)
 - ▶ Newest technique: HYSPLIT dispersion model (NOAA)

Existing Work to Date

- ▶ Consensus that there are large negative effects in developing countries
 - ▶ Little evidence on why
- ▶ Open questions:
 - ▶ Are marginal damages higher or lower in LMICs? (Colmer et al.,2021)
 - ▶ Are differences in marginal damages because of non-linear dose-response
 - ▶ Are differences in marginal damages because of weak institutions/development?
 - ▶ Long term effects of exposure? Consumption smoothing?
- ▶ Conceptual frameworks needed

Lets formalize our intuition

- ▶ Weather can harm health directly (physiological stress)
 - ▶ Or indirectly, e.g. by lowering incomes of those in weather-dependent (outdoor) jobs
- ▶ There is a market for defensive goods (e.g. medicine, AC, etc.) which $\uparrow P(\text{survival})$
- ▶ But in developing countries, credit constraints restrict consumption smoothing
- ▶ Lets incorporate this into a model and study the comparative statics
 - ▶ Goal: explain how weather affects health, considering financial constraints
 - ▶ Health is endogenously determined by consumption and health investments

Setup

- ▶ **Agents:** value consumption c_{it} and health h_{it} .
- ▶ **Objective:** Maximize lifetime utility

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u_i(c_{it}, h_{it}) \right]$$

- ▶ **Health Status:** Determined by health investments (q_{it}) + exogenous weather (z_{it})

$$h_{it} = h_i(\underbrace{q_{it}}_{(+)}, \underbrace{z_{it}}_{(-)})$$

Production

- ▶ c_{it} produced by $f(\cdot)$, which requires capital + labor
- ▶ Defensive good q_{it} traded at price P_t

Labor

- ▶ Agent endowed with labor a_{it} (random), sold at wage ω_t
- ▶ Labor can be affected by weather (z_{it})

$$a_{it} = a_i(z_{it})$$

- ▶ Agent can (partially) save/borrow to (partially) smooth consumption over t
- ▶ Add financial market frictions characteristic of LMICs
- ▶ **Financial Constraints:** Savings (x_{it}) and borrowing are limited by:
 - ▶ Borrowing constraint (minimum savings): $x_{it} \geq -b_i(r_t)$
 - ▶ Saving constraint (maximum savings): $x_{it} \leq b_i(r_t)$
 - ▶ $b(\cdot)$ unrestricted s.t. agent can be unconstrained
- ▶ **Budget Constraint:** income is used for consumption and health investments:

$$x_{i,t+1} \leq r_t x_{it} + w_t a_i(z_{it}) - P_t q_{it} - c_{it}$$

Equilibrium

- ▶ All markets must clear each t , which endogenously determines prices (r_t, w_t)
 - ▶ Note P_t is exogenously set in world market
- ▶ Returns to capital/labor are functions of their aggregate
 - ▶ Capital demand: from production function $f(\cdot)$
 - ▶ Capital supply: savings transformed into capital for production
- ▶ **Market Equilibrium:** set optimal demand for labor/capital = aggregate supply
 - ▶ where $K_t = \int x_{it}^* di$
 - ▶ interest rates r_t are determined in equilibrium as:
$$r_t = r(K_t)$$
 - ▶ **same equilibrium logic for wages**

Mechanisms relating weather to health

- ▶ **Direct health effect:** random shock z_{it} enters health directly: $h_{it} = h_i(q_{it}, z_{it})$
 - ▶ Physiological channel: extreme temperature stresses cardiovascular/respiratory system
 - ▶ Impact can depend on region, $h_i(\cdot)$
- ▶ **Indirect health effect:** through productivity $a_i(z_{it})$
 - ▶ Likely in rural areas, where ag workers are outside
- ▶ **Mitigation:** agent minimizes neg. utility shock from z_{it} by purchasing q_{it} .
 - ▶ This could also be opp cost of working inside (at lower w_{it}) to avoid outdoors
 - ▶ Effect of mitigation limited by $\frac{\partial h_i(q_{it}, z_{it})}{\partial q_{it}}$
- ▶ **Smoothing:** agent can also mitigate by reallocating resources b/w periods
 - ▶ Agents with near-binding borrowing constraints experience worse shocks

Comparative Statics

- ▶ How can we go from household model to aggregate statements about weather-health?
- ▶ **Object of interest:** aggregate covariance b/w health h_{it} and weather z_{it} :

$$H_t = \left| \frac{\int h_{it} z_{it}^{(T)} di}{\int h_{it} di} \right|$$

- ▶ **Rural Areas:** Consider a set of changes that:
 - ▶ $\uparrow$ responsiveness of $a_i(z_{it})$ to z_{it}^T (more exposed to weather extremes).
 - ▶ $\uparrow$ prices for health goods P_t above that of urban (lower market access)
 - ▶ $\downarrow$ incomes by reducing $a_i(z_{it})$ for any z_{it}
 - ▶ Any one of these leads to increase in H_t

Prediction: rural areas will have stronger correlation b/w weather and health

Effect of Bank Access

- ▶ **Relaxing financial constraints** increases either or both of $(-b_i(r_t), b_i(r_t))$
 - ▶ Partial credit modeled as increase in $b_i(r_t)$ only (i.e. can save more)
 - ▶ Allows agents to smooth consumption; draw down savings to mitigate weather shock
 - ▶ Reduces the correlation between weather shocks z_{it} and health outcomes h_{it} .

- ▶ **Prediction:** Prediction: $H_t \downarrow$ as $b_i(r_t) \uparrow$
 - ▶ Areas with better financial access have smaller effect of z_{it} on health

Deschenes et al. (2013): Weather and Death in India

- ▶ **Research question:** how do temperature shocks affect mortality?
 - ▶ How does that differ between urban/rural areas?
 - ▶ What is the role of financial market frictions as an intermediary?
- ▶ **Setting:** Indian districts from 1957-2000
- ▶ **Data:** daily gridded weather + annual infant mortality
- ▶ **Design:** panel fixed effects
 - ▶ Difference-in-differences with rural bank expansion
- ▶ **Result:** 1) heat $\uparrow$ mortality, 2) driven by rural populations, 3) credit access mitigates effect by facilitating consumption smoothing

Aside: Climate change versus weather

- ▶ Climate change is long term, weather is short term
- ▶ Studying impacts of weather is easier:
 - ▶ Usually use high dimensional fixed effects
 - ▶ E.g. with district and state-year FEs, exploit monthly deviations from average
- ▶ Studying impacts of climate is harder:
 - ▶ There is no good counterfactual for climate change

Productivity Data

Rural

- ▶ Yields: annual, district level output/cultivated area (27 crops) (World Bank)
- ▶ Ag price index: mean crop price weighted by value in district (World Bank)
- ▶ Ag wages: daily wage rate (World Bank)

Urban

- ▶ Manufacturing productivity: output per worker (ASI)
- ▶ Urban CPI: NSS survey
- ▶ Manufacturing wages: $ASI \text{ (nominal wage / CPI)}$

Flexible specification to model effect of daily temperature:

$$Y_{dt} = \sum_{j=1}^{11} \theta_j TMEAN_{dtj} + \sum_k \delta_k 1\{RAIN_{dt} \text{ in tercile } k\} + \alpha_d + \gamma_t + \lambda_r^1 t + \lambda_r^2 t^2 + \epsilon_{dt}$$

- ▶ r is a climatic region
- ▶ $TMEAN$ is days in year t where:
 - ▶ daily mean temp $< 70^\circ$, $\geq 97^\circ$, or in the nine 3° bins in between
- ▶ θ_j 's capture non-linearities, despite having annual unit of analysis
- ▶ $RAIN$ accounts for correlation b/w temperature and rainfall

Identifying Assumptions?

Secondary Approach

- ▶ Consolidates same information
- ▶ Workhorse model nowadays because more parsimonious

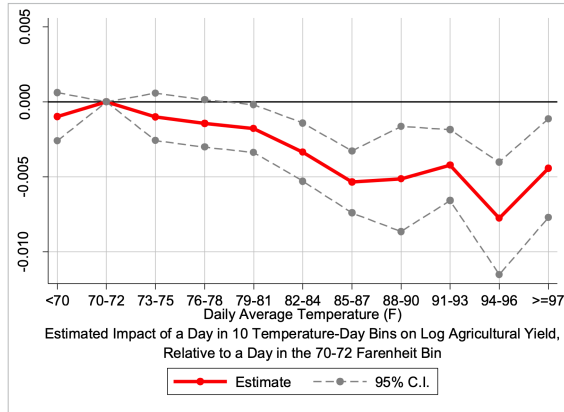
$$Y_{dt} = \beta CDD80_{dt} + \sum_{k=1}^3 \delta_k 1\{RAIN_{dt} \text{ in tercile } k\} + \alpha_d + \gamma_t + \lambda_r^1 t + \lambda_r^2 t^2 + \epsilon_{dt}$$

- ▶ where CDD80 is cumulative degree days in district that exceeded 80°. e.g. if two days over 80, one at 82 and other at 84, then CDD=2+4=6
- ▶ Special version of binned regression:
 - ▶ All bins $j < 80^\circ$ are zero
 - ▶ $j > 80^\circ$ bins restricted to be linearly increasing
- ▶ Assumption: days when mean daily temp below 80 irrelevant for determining Y

Results: Yields

Test of indirect channel through labor productivity, $a_i(z_{it})$

Figure 3: The Effect of Daily Average Temperatures on Log Agricultural Yields.

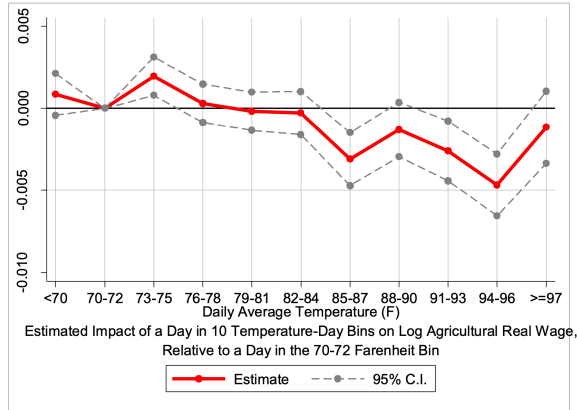


A single day with temperature over 85° reduces yields by 0.5%

Results: Wages

Test of indirect channel through $a_i(z_{it})$

Figure 4: The Effect of Daily Average Temperatures on Log Real Agricultural Wages.



A single day with temperature over 85° reduces ag wages by 0.25-0.5%

Results: Urban vs. Rural

Table 2: Weather and Incomes - Rural-Urban Differences, Exposure by Calendar Year				
Dependent Variable:	Rural		Urban	
	Log (Productivity)	Log (Real Wages)	Log (Productivity)	Log (Real Wages)
	(1)	(2)	(3)	(4)
A. Temperature (degree-days over 90F)/10	-0.0023*	-0.0023***	0.0003	0.0012
(std error)	(0.0009)	(0.0006)	(0.0033)	(0.0036)
[Effect of 1 std dev in CDD90]	[-0.023]	[-0.022]	[0.002]	[0.009]
B. Temperature (degree-days over 80F)/10	-0.0034***	-0.0026***	-0.0006	0.0006
(std error)	(0.0006)	(0.0004)	(0.0013)	(0.0013)
[Effect of 1 std dev in CDD80]	[-0.126]	[-0.098]	[-0.021]	[0.021]
C. Temperature (degree-days over 70F)/10	-0.0023***	-0.0018***	-0.0010	0.0005
(std error)	(0.0004)	(0.0003)	(0.0008)	(0.0008)
[Effect of 1 std dev in CDD70]	[-0.132]	[-0.101]	[-0.053]	[0.023]
Observations	8,304	8,304	512	592

► This corroborates Prediction 1 from model!

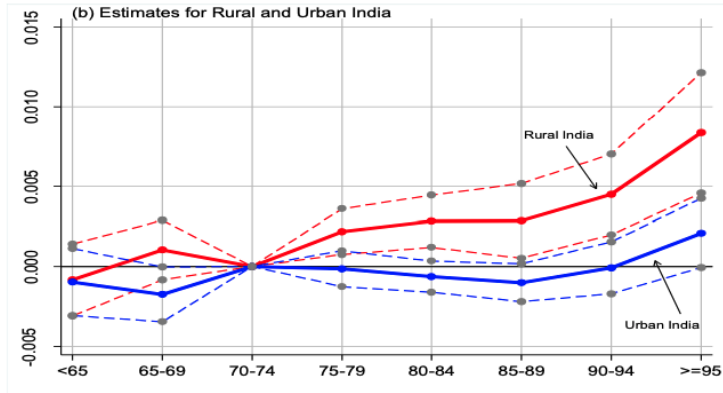
Results: by Growing Season

Table 2: Impact of Daily Temperature on All-Age Mortality in Rural and Urban India.

	Rural		Urban	
	Temperature		Temperature	
	75-89°F (1a)	>90°F (1b)	75-89°F (2a)	>90°F (2b)
A. Exposure over Calendar Year				
Impact of Temperature	0.0021**	0.0047***	0.0001	0.0012
(std error)	(0.0007)	(0.0010)	(0.0005)	(0.0008)
B. Exposure over Agricultural Calendar				
Impact of Growing Season Temperature	0.0028***	0.0067***	0.0005	0.0016
(std error)	(0.0008)	(0.0016)	(0.0006)	(0.0013)
Impact of Non-Growing Season Temperature	-0.0047*	-0.0024	-0.0032	-0.0020
(std error)	(0.0023)	(0.0023)	(0.0017)	(0.0016)
Test of equality across agricultural seasons	0.002	0.001	0.058	0.060

► This is also consistent with Prediction 1 from model!

Results: Mortality



Rural: years with more days with exceeding 87° have 0.5% higher mortality

Bank Access

- ▶ Prediction 2: bank access enables consumption smoothing
 - ▶ Relaxing borrowing/saving constraints should mute relationship
- ▶ Test: Between 1977-1990, central bank extends banking to the poor
 - ▶ Existing banked (urban) areas must open 4 branches in unbanked (rural) areas
- ▶ Method: difference in differences
 - ▶ Compare mortality in hot and cold places before and after bank access
 - ▶ Issues with this design?

Results: Credit Access

	Panel Estimates, Rural		2SLS, Rural		2SLS, Urban	
	Temperature		Temperature			
	75-89 °F	>90 °F	75-89 °F	>90 °F	75-89 °F	>90 °F
	(1a)	(1b)	(2a)	(2b)	(3a)	(3b)
Dependent Variable is Log Annual Mortality Rate						
Main Effect of Temperature (std error)	0.0024*** (0.0009)	0.0067*** (0.0012)	0.0026*** (0.0010)	0.0121*** (0.0019)	0.0000 (0.0007)	0.0017 (0.0021)
Main Effect of Number of Bank Branches (std error)			1.0633 (0.7407)		0.1262 (0.1143)	
Temperature x Bank Branches (std error)			-0.0041* (0.0021)	-0.0216*** (0.0055)	-0.0010 (0.0006)	-0.0013 (0.0014)

- ▶ Effect of moderate and hot days on mortality is moderated by bank access
- ▶ Bank branches more than offset weather-death relationship

Takeaway: credit enables poor to withstand environmental shocks

- ▶ Aggregate data entangles ambient means and experienced exposure
- ▶ What exactly makes rural areas more vulnerable?
- ▶ How much of the weather-death effect is direct vs. indirect?
 - ▶ Empirically, we did not identify model parameters separately
- ▶ Will weather shock be enough to affect remaining lifetime income?

Way Forward

- ▶ Bank access represents anything that relaxes budget constraint
- ▶ Policy could have been subsidies on q_{it} , encouraging seasonal migration, etc.
 - ▶ Open areas for research

RCT Evidence: Cash Transfers and Adaptation to Extreme Heat



Speaker

Sheetal Sekhri, University of Virginia

When: Friday, September 25, from 10:30 - Noon (Central Time)

Where: Ruttan Hall 119 & [Zoom](#)

Title

Forecast-Triggered Cash Transfers and Adaptation to Extreme Heat: Evidence from Gig Workers in India

Abstract

We study whether short-term liquidity relaxes frictions that prevent adaptation to extreme heat. In a randomized intervention, gig delivery workers received small, forecast-triggered payments during heatwaves. Access to liquidity attenuates declines in labor supply and productivity and mitigates heat-related symptoms by enabling costly avoidance behaviors, including rescheduling work to cooler hours and resting in cooled locations. Machine-learning heterogeneity analysis shows that gains are concentrated among workers facing rigid demand and output-linked pay. Despite substantial ex post benefits, workers exhibit low willingness to pay for equivalent ex ante heat-indexed insurance, highlighting the importance of timing in climate adaptation.

Back to the main question

Back to the question

- ▶ How does environmental quality affect economic development?
- ▶ Most reduced form studies identify $\frac{\partial h}{\partial e}$
- ▶ Recall, this is an input into estimating benefits (MTWP) of $\uparrow e$
 - ▶ which we need to know in order to set optimal environmental policy
 - ▶ optimal in the sense that agents decisions are first-best
- ▶ Is physical health the only thing that matters for estimating $\frac{\partial h}{\partial e}$?

Chen et al. (2024): Air Pollution and Mental Health

- ▶ **Research question:** What is the impact of air pollution on mental health?
 - ▶ Direct mechanism: oxidative stress
 - ▶ Indirect mechanism: labor productivity, income, physical activity
- ▶ **Setting:** China in 2014-2015
- ▶ **Data:** Kessler Psychological Distress Scale
 - ▶ PM2.5 from China National Environmental Monitoring Center
- ▶ **Design:** instrument pollution with thermal inversions
- ▶ **Results:** Pollution $\uparrow$ depression
 - ▶ Mechanisms: lower exercise, worse physical health
 - ▶ Total cost = \$22.88 billion, similar to mortality costs!

Background

- ▶ Why do we care about mental health?
 - ▶ Poverty associated with life satisfaction and **stress**
 - ▶ These mechanisms affect productivity, which perpetuates poverty
 - ▶ See Haushofer and Salicath (2023); Haushofer and Fehr (2014)
- ▶ Frontier in development economics: RCTs on income and stress
 - ▶ Cash transfers ↑ subjective well-being (McGuire et al., 2022)
 - ▶ Cash transfers ↓ suicide in Indonesia (Christian et al., 2019)
 - ▶ Cash transfers ↓ depression (Haushofer and Shapiro, 2016)
- ▶ Toxicology literature: pollution ↑ oxidative stress (Calderon-Garciduenas et al., 2003; Sørensen et al., 2003; MohanKumar et al., 2008, Power et al., 2015)

- ▶ Theory of change: **pollution** → **stress** → productivity
 - ▶ Poverty correlated with pollution and stress
 - ▶ Direct effect of pollution on productivity (Graff Zivin & Niedell, 2012; California)
 - ▶ Many parts of the chain are open areas for research!

Air Pollution and Mental Health

What are the empirical challenges?

① OVB: pollution correlated with economic activity

- ▶ Richer counties more polluted
- ▶ Induces **negative** bias
- ▶ Solution: ?

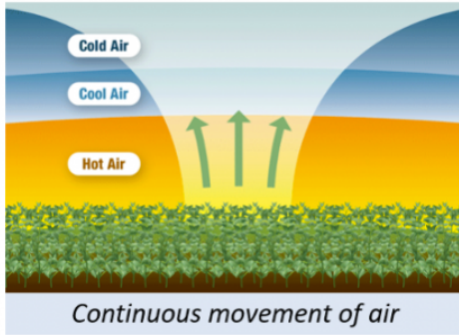
② ?

③ ?

④ ?

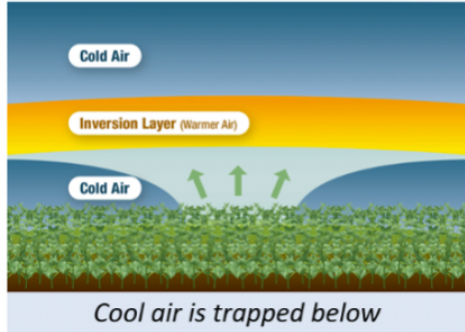
Instrumental Variable: Thermal Inversion

During a typical day



Agweb.com

With an inversion



Agweb.com

Instrumental Variable: Thermal Inversion



Instrumental Variable: Thermal Inversion

- ▶ Air is warmer near surface, becomes cooler at higher altitudes
- ▶ Thermal inversion: temperature-altitude gradient is reversed
 - ▶ Rare phenomenon
 - ▶ Depends on **random** meteorological factors
- ▶ Cool air is denser and does not rise
- ▶ Pollution becomes trapped near surface

What do you think?

- ▶ Mental health: 2014 China Family Panel Study Survey
 - ▶ 16,000 rural and 13,000 urban residents (162 counties out of 1,355)
 - ▶ Six questions about mental health on 0-24 Scale
 - ▶ Depression, nervousness, restlessness, hopelessness, effort, worthlessness
- ▶ Pollution: PM2.5
 - ▶ Webscraped
 - ▶ Validate with satellite data?
- ▶ Thermal inversion (MERRA-2 Air Temperature Layers)
 - ▶ Inversion = 1 if temperature of first layer (110m) < second layer (320m)
 - ▶ IV: number of inversions in month prior to each interview

Inversions and Pollution

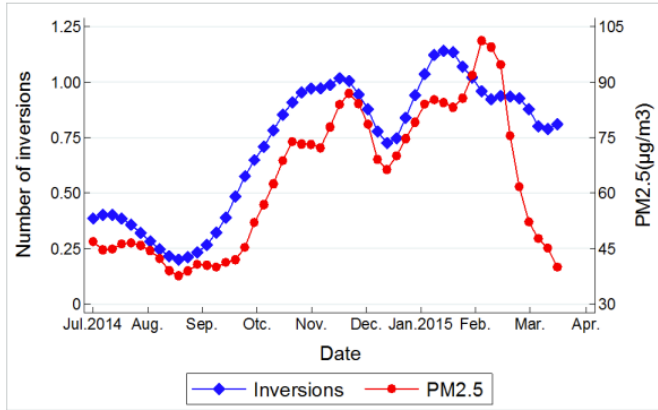


Figure 1. Time Trend of PM_{2.5} and Thermal Inversion

Do thermal inversions seem rare and random?

First Stage:

$$P_{ict} = \alpha_0 + \alpha_1 I_{ict} + f(W_{ict}) + \gamma_c + \sigma_t + \mu_{ict}$$

Second Stage:

$$H_{ict} = \beta_0 + \beta_1 P_{ict} + f(W_{ict}) + \gamma_c + \sigma_t + \epsilon_{ict}$$

- ▶ P_{ict} is PM2.5 in month prior to interview
- ▶ I_{ict} is instrument, number of thermal inversions in past month
- ▶ H_{ict} is total mental illness score (0-24)
- ▶ $f(W_{ict})$ are flexible weather controls
- ▶ σ_t are year-month FEs (**important!**)

Main Results: Mental Health

TABLE 1—FIRST-STAGE AND IV RESULTS

Dep. var.	First stage		Second-stage estimation			
	PM _{2.5}		K6 score		Severe mental illness	
	(1)	(2)	(3)	(4)	(5)	(6)
Thermal inversions	0.2118 (0.0469)	0.2638 (0.0357)				
PM _{2.5}			0.1229 (0.0500)	0.1164 (0.0388)	0.0047 (0.0025)	0.0042 (0.0019)
Mean [SD] of dep. var.	47.7132	[18.0450]	2.9556	[3.7598]	0.0438	[0.2047]
Weather controls	No	Yes	No	Yes	No	Yes
KP <i>F</i> -statistics	20.36	54.56				

- ▶ 1 μ/m^3 PM_{2.5} ↑ mental illness by 0.1164 units (4% of mean; 0.4 σ)
- ▶ 1 μ/m^3 PM_{2.5} ↑ prob. of several mental illness by 0.42pp (9.5% of mean)

Main Results: Symptoms

Table 4. Effect of Air Pollution on Mental Health: By Symptom

	About how often do you feel					
	Depression (1)	Nervousness (2)	Restlessness (3)	Hopelessness (4)	Difficulty (5)	Worthlessness (6)
PM _{2.5}	0.0155** (0.0073)	0.0045 (0.0062)	0.0134*** (0.0044)	0.0150*** (0.0048)	0.0183*** (0.0055)	0.0123** (0.0049)
Observations	12,657	12,659	12,660	12,638	12,657	12,649
County <i>FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
Year-by-month <i>FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample weights	Yes	Yes	Yes	Yes	Yes	Yes
KP <i>F</i> -statistic	36.17	36.19	36.20	36.11	36.15	36.09

- Not driven by any one particular symptom

Main Results: Persistence

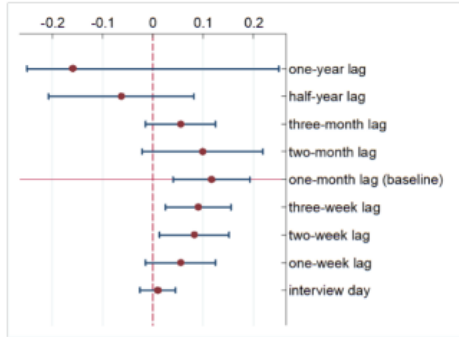


Figure A.1: Impact of PM2.5 on the K6 score

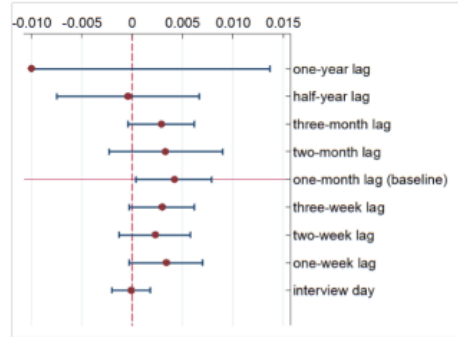


Figure A.2: Impact of PM2.5 on K6 ≥ 12

- Effect of pollution exposure on mental health dissipates over time

- Focus on indirect effects (physical health, productivity, behavior)

Table 6. Mechanism Tests

	Exercise (past 1 week)			Sickness (past 2 weeks)		Self-rated health (past 1 month)
	(1)	(2)	(3)	(4)	(5)	(6)
	1=yes, 0=no	times	hours	1=yes, 0=no	1=not serious, 5=very serious	1=very healthy, 5=very unhealthy
PM _{2.5}	-0.0049** (0.0022)	-0.0068* (0.0035)	-0.0448** (0.0209)	0.0020 (0.0019)	0.0089* (0.0052)	0.0092* (0.0056)
Observations	12,664	12,663	12,670	12,670	3,806	12,668
Mean of Dep. Var.	0.4670	2.4008	3.3681	0.2917	3.1140	2.9698
S.D. of Dep. Var.	0.4989	3.1324	6.4268	0.4545	1.4053	1.1816
County <i>FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
Year-by-month <i>FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes	Yes	Yes
Sample weights	Yes	Yes	Yes	Yes	Yes	Yes
KP <i>F</i> -statistic	48.73	48.57	48.72	44.92	59.36	36.30

- Driven by reduction in physical exercise, not other illness
- Very small impact compared to pollution effect (0.002σ vs. 0.4σ)

Heterogeneity

	The K6 Score			
	(1)	(2)	(3)	(4)
	Male	Female	Age 16-60	Age >=60
PM _{2.5}	0.0986*** (0.0268)	0.0575* (0.0342)	0.0611** (0.0298)	0.1499** (0.0599)
Observations	6,093	6,522	9,446	3,169
Mean of Dep. Var.	2.6378	3.2799	2.9504	3.0281
S.D. of Dep. Var.	3.5378	3.9405	3.5819	4.2683
County FE	Yes	Yes	Yes	Yes
Year-by-month FE	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes
Sample weights	Yes	Yes	Yes	Yes
KP <i>F</i> -statistic	25.14	43.22	32.11	27.20

- Why are effects larger for men?

Economic Cost? Back of the Envelope

- ▶ 1σ ↑ in PM2.5 leads to 9.5% ↑ in prob. of severe mental illness
 - ▶ About 0.095×1.4 billion = 13.3 million additional people
- ▶ Annual cost per patient of mental illness in China is \$USD 3,665 (Xu et al 2016)
- ▶ Assume 8% of patients with mental illness get treated (Phillips et al., 2009)
- ▶ **Economic cost of mental illness:** $13.3 \times 3665 \times 0.008 \times 12 = 46.7$ billion USD
- ▶ Mortality cost of 1σ PM2.5 is \$USD 30 billion in USA (Deryugina et al., 2016)

Issues with the back of the envelope calculation?

Discussion

- ▶ productivity and direct income effects

Lecture 5

How does the environment affect economic development?

Housekeeping

- ▶ See last week's slides on pollution and mental health (Chen, Oliva & Zhang, 2024)
- ▶ Monday paper: no need to fully understand WTP estimation
- ▶ Monday lecture: print the slides and bring to class!
- ▶ Research Proposal: first draft due in 12 days
- ▶ Decide on a replication paper

- ▶ **Guiding question:** how does the environment affect economic development?

$$MWTP_e = \frac{\lambda_e}{\lambda_y} = \frac{1}{\lambda_y} \left(\frac{\partial u}{\partial e} + \frac{\partial u}{\partial h} \frac{\partial h}{\partial e} \right) + \frac{\partial \Delta y}{\partial e} + \frac{\partial \Delta y}{\partial h} \frac{\partial h}{\partial e}$$

- ▶ Today's focus: productivity and income
 - ▶ Indirect: worker productivity (lots of research)
 - ▶ Direct: income (very little research)

Environment, Productivity, and Income

- ▶ Temperature is neg. associated w/ aggregate output (Dell, Jones, and Olken 2012)
 - ▶ Specifically in poor countries
 - ▶ Driven by agriculture and manufacturing sectors
 - ▶ Effects on levels (crop output/yields) and growth rates
- ▶ Evidence on direct channel: high temperatures associated w/ lower crop yields
 - ▶ Mendelsohn and Dinar (1999), Auffhammer et al. (2006), Schlenker and Roberts (2009), Lobell et al. (2011), and Gupta et al. (2017)
 - ▶ Very little evidence on other direct channels (e.g. ecotourism)
- ▶ Evidence on indirect channels: what we don't know
 - ▶ **conflict, institutions, political stability**
 - ▶ open areas for research — see Burke, Hsiang, and Miguel (2015)
- ▶ Evidence on indirect channels: what we do know
 - ▶ crop/worker productivity: see reading list

Somanathan et al. (2021): Temperature, Productivity, and Labor

- ▶ **Research question:** What is the effect of temperature on firm output?
 - ▶ How does heat stress mediate the temperature-output relationship
- ▶ **Setting:** India's manufacturing sector (cloth weaving, sewing, steel)
- ▶ **Data:** High-frequency (15 year panel, daily)
 - ▶ Worker-level, sewing line level, team level, plant-year, district-year
- ▶ **Design:** panel fixed effects
- ▶ **Result:** 1) heat ↓ plant-level annual output, 2) driven by output elasticity of labor

Background

- ▶ Workers should respond quickly when working in uncomfy temperatures
 - ▶ Heat impacts on labor can be identified from daily or aggregated data
 - ▶ Distinguishes heat stress from alternative mechanisms which materialize over long time scales (i.e. conflict)
- ▶ Two channels through which temperatures affect workers
 - ▶ Produce less (productivity)
 - ▶ Absenteeism
- ▶ Policy: is cost-effective to invest in climate control in dev countries?
 - ▶ Depends on value-added per worker
 - ▶ If too low, then electricity cost not worth it

Data: Worker + Cloth Weaving

- ▶ Worker data
 - ▶ Surveyed **output and attendance** from cloth-weaving, garment, and steel firms
 - ▶ Three cloth-weaving factories in Surat,
 - ▶ Eight garment sewing plants (Delhi, Hyderabad, Chhindwara)
 - ▶ Steel plant (one large plant in Central India)

- ▶ Cloth data (2012): 147 workers
 - ▶ Daily meters woven (for each worker)
 - ▶ Worker pay slips each day (output + attendance)
 - ▶ No climate control

Data: Garment Sewing

- ▶ Garment sewing (2012-2013): Line level
 - ▶ Lines of 10-20 workers
 - ▶ Productivity: hourly output relative to target
 - ▶ Daily production from 103 sewing lines
- ▶ Climate control
 - ▶ Phased rollout of cooling equipment
 - ▶ 84 of 103 sewing lines in Delhi, of which 74 had cooling

Data: Rail and Steel Plant

- ▶ Supplies rectangular blocks of steel to Indian Railways
- ▶ Three shifts per day
- ▶ Workers assigned to one of three teams, which rotate across shifts
- ▶ Data:
 - ▶ Team and number of blocks for each shift during 1999-2008
 - ▶ 9,172 shifts over 3,337 working days
 - ▶ Team-level absences over 857 days for 2000-2003

Garment factory and cloth weaving



Rail/Steel Mill



Data: Panel

- ▶ Annual Survey of Industries (1999-2013)
 - ▶ Census of large manufacturing plants (> 100 workers)
 - ▶ Random sample of small plants
 - ▶ Output, fixed assets, input expenses, employment
 - ▶ Unbalanced panel of 58,377 plants
- ▶ District Manufacturing GDP (1998-2009)
 - ▶ Planning Commission of India
 - ▶ ASI plants + unregistered plants
- ▶ Weather: public weather stations + gridded data (IMD)
 - ▶ Daily maximum at closest station to factory
 - ▶ Gridded data (1 degree) for panel data

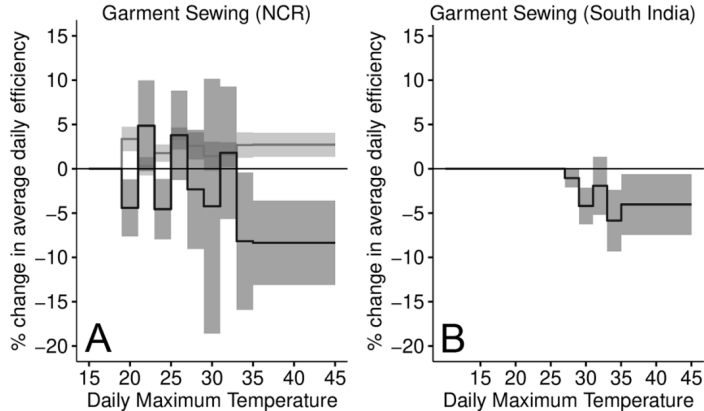
Temperature-output estimated with weekly data as follows:

$$y_{iw} = \alpha_i + \gamma_M + \gamma_t + \sum_{j=2}^J \beta_j T_{iw}^j + \theta R_{iw} + \lambda X_{iw} + \epsilon_{iw}$$

- ▶ y_{iw} is physical output by worker i in week w
- ▶ If absent, output=0 for that day before aggregating to week
- ▶ α_i is individual FE (what does it account for?)
- ▶ $\gamma_{M/t}$ are month/year FEs (what does it account for?)
- ▶ T_j counts days falling into temperature bin j (captures nonlinearities)

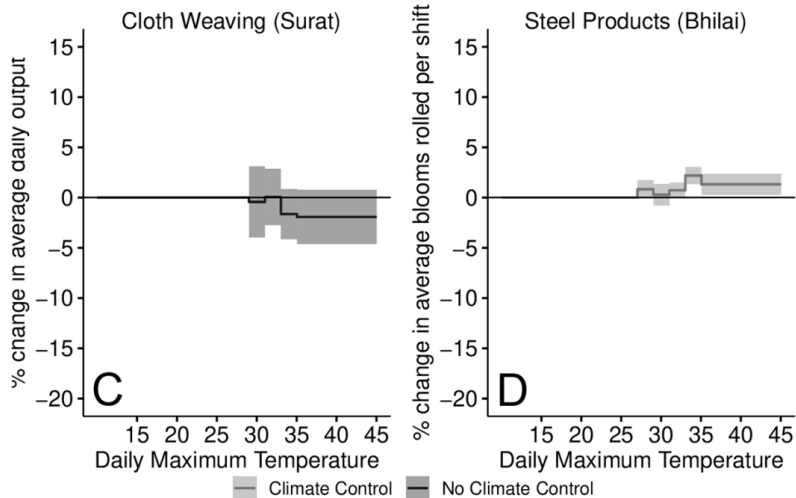
Identification Assumptions

Results: Garment Sewing



- ▶ output is lower in hotter weeks **without climate control**
- ▶ Moving from lowest to highest temperature bin ↓ productivity by 8%
- ▶ Note: these are 90% CI's!

Results: Cloth and Steel



Lagged Effects

- ▶ Previously, β_j combined productivity and absenteeism channel
- ▶ Now we use daily data

$$y_{id} = \alpha_i + \gamma_M + \gamma_t + \sum_j \beta_j T_{id}^j + \sum_j \omega_j L_{id}^j + \theta R_{id} + \lambda X_{id} + \epsilon_{id}$$

- ▶ where T^j is now an indicator for day d falling into temperature bin j
- ▶ X_{id} includes day-of-week FE
- ▶ L_{id}^j counts number of days in bin j during six preceding days

- ▶ Is lag specification plausible?
 - ▶ Is six days a plausible window to detect cumulative exposure?
- ▶ What are the identification assumptions?
- ▶ What are the threats?

Lagged Effects: Productivity

TABLE 2
EFFECT OF HOT DAYS ON WORKER OUTPUT

	CLIMATE CONTROL		NO CLIMATE CONTROL		
	Garments (Log Efficiency) (1)	Steel (Log Blooms Rolled) (2)	Weaving (IHS Meters) (3)	Garments (Log Efficiency) (4) (5)	
<i>T</i> (33°–35°C)	.025** (.010)	.028* (.017)	–.040** (.019)	–.129*** (.042)	–.007 (.037)
<i>T</i> (above 35°C)	.035*** (.014)	.020** (.009)	.011 (.022)	–.154*** (.041)	.008 (.046)
<i>L</i> (33°–35°C)	–.004 (.005)	.005 (.004)	–.033*** (.011)	–.009 (.012)	.004 (.010)
<i>L</i> (above 35°C)	–.011** (.005)	–.002 (.005)	–.027*** (.009)	–.019 (.027)	.015 (.018)
Climate control	Yes	Yes	No	No	No
Number of units	74 lines	9 teams	147 workers	10 lines	19 lines
Time span (days)	730	3,337	365	730	730

- ▶ Lower output on hot days in sites w/o climate control
- ▶ Weavers: additional day $> 35^{\circ}$ in past week $\downarrow$ daily output by 3%

Lagged Effects: Absenteeism

TABLE 3
EFFECT OF HOT DAYS ON WORKER ABSENTEEISM

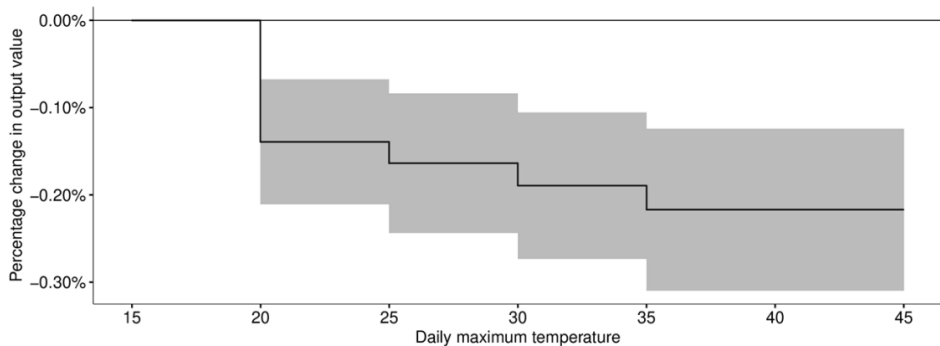
	CLIMATE CONTROL			NO CLIMATE CONTROL		
	Garments		Steel (All) (3)	Weaving (All) (4)	Garments	
	Paid (1)	Unpaid (2)			Paid (5)	Unpaid (6)
<i>T</i> (33°–35°C)	.082*** (.022)	–.083 (.065)	–.011 (.048)	.003 (.004)	–.001 (.128)	.796 (.678)
<i>T</i> (above 35°C)	.115*** (.027)	.031 (.049)	.051 (.068)	–.004 (.004)	–.034 (.117)	1.001 (.862)
<i>L</i> (33°–35°C)	–.018 (.011)	–.047 (.032)	.044*** (.014)	.006*** (.002)	.017 (.077)	.772 (.686)
<i>L</i> (above 35°C)	.021** (.010)	–.001 (.022)	.045** (.020)	.005*** (.002)	.078 (.083)	.567 (.426)
Number of units	224 lines		9 teams	147 workers	42 lines	
Time span (days)	730		3,337	365	730	

- How do you interpret these results?
- new research idea: effect of paid leave on temperature-productivity relation

Annual Survey of Industries: 15-year panel results

$$y_{it} = \alpha_i + \gamma_t + \sum_{j=2}^5 \beta_j T_{it}^j + \theta R_{it} + \epsilon_{it}$$

outcome is log value of plant i output in year t



Recap of Results

- ▶ Daily worker data: workers are less productive on hot days
- ▶ Daily worker data: (kind of) more absenteeism on hot days
- ▶ Plant panel: **aggregate** output is lower on hot days
 - ▶ This is the object we are interested in
 - ▶ Remember $\frac{\partial \Delta y}{\partial e}$ and $\frac{\partial \Delta y}{\partial h} \frac{\partial h}{\partial e}$
- ▶ **How much of the temperature-induced ΔY is due to ΔMP_L ?**
 - ▶ As opposed to changes in capital intensity
 - ▶ No data on climate control in ASI data

Lets make a Cobb-Douglas model

Consider a production function:

$$Y = A \cdot K^{\omega} \cdot L^{\beta}$$

Now modify by letting elasticities be a function of temperature, T

$$Y = A(T) \cdot K^{\omega(T)} \cdot L^{\beta(T)}$$

where linearity is assumed:

$$A(T) = A_0 + \sum a_j T_j$$

$$\omega(T) = \omega_0 + \sum \omega_j T_j$$

$$\beta(T) = \beta_0 + \sum \beta_j T_j$$

Estimating Equation

- Log-linearizing, we get Equation (7) from the paper:

$$y = \alpha_0 + \underbrace{\sum_{j=2}^5 \alpha_j T^j}_{\text{TFP effect}} + \omega_0 \cdot k + \underbrace{\sum_{j=2}^5 \omega_j T^j k}_{\text{effect on capital elasticity}} + \beta_0 \cdot l + \underbrace{\sum_{j=2}^5 \beta_j T^j l}_{\text{Effect on labor elasticity}}$$

- Data:
 - k: opening value of equipment and machinery at start of financial year
 - l: number of full-time workers

Results

- ▶ Temperature effects on output elasticity of L is negative and significant
 - ▶ i.e. temperature makes labor less productive at higher temperatures
- ▶ Temperature effects on output elasticity of K are (weakly) positive
- ▶ Controlling for interactions, residual temperature-output effect is zero
- ▶ Since overall temperature-output relationship is negative, temperature-induced declines in labor productivity drive the effect

Results

TEMPERATURE INTERACTIONS WITH FACTOR INPUTS

	(1)	(2)	(3)	(4)	(5)
T^a					.02324* (.01345)
T^2	.00256** (.00097)	.00008 (.00211)	-.00008 (.00162)		
T^3	.00147 (.00103)	-.00205 (.00229)	-.00009 (.00165)		
T^4	.00081 (.00108)	-.00094 (.00237)	-.00028 (.00170)		
T^5	.00003 (.00118)	-.00499* (.00259)	-.00171 (.00185)		
l	.8612*** (.0957)		.91426*** (.09660)	.36520*** (.05910)	
k		.20433** (.05674)	.06629 (.04114)		
$l \times T^2$	-.00098*** (.00027)		-.00134*** (.00034)	-.00056** (.00022)	
$l \times T^3$	-.00067** (.00027)		-.00104*** (.00027)	-.00038** (.00017)	
$l \times T^4$	-.00052* (.00027)		-.00077*** (.00027)	-.00030* (.00017)	
$l \times T^5$	-.00036 (.00029)		-.00075*** (.00029)	-.00039** (.00018)	
$k \times T^2$		-.00009 (.00015)	.00028* (.00015)		
$k \times T^3$		.00005 (.00016)	.00022* (.00012)		
$k \times T^4$		-.00003 (.00016)	.00016 (.00011)		
$k \times T^5$		.00022 (.00018)	.00024** (.00012)		

- ▶ Identification
- ▶ Did they do everything they could with the data?
- ▶ Alternative explanations

Back to the model

$$MWTP_e = \frac{\lambda_e}{\lambda_y} = \frac{1}{\lambda_y} \left(\frac{\partial u}{\partial e} + \frac{\partial u}{\partial h} \frac{\partial h}{\partial e} \right) + \frac{\partial \Delta y}{\partial e} + \frac{\partial \Delta y}{\partial h} \frac{\partial h}{\partial e}$$

- ▶ Most research in environment/development economics seeks to identify $\frac{\partial h}{\partial e}$
 - ▶ At this point, $\frac{\partial h}{\partial e} > 0$ is unambiguous
 - ▶ Focus is on **magnitudes**
- ▶ What about indirect effects via productivity $(\frac{\partial \Delta y}{\partial h} \frac{\partial h}{\partial e})$?
 - ▶ This is the research frontier
- ▶ Next: what about $\frac{\partial \Delta y}{\partial e}$, the direct effect of environment on incomes
 - ▶ Almost no research on this

How might the environment **directly** affect incomes?

Conceptual Framework

- ▶ Assume landholders vary in distance to a central market
- ▶ Each chooses land and labor to maximize farm profits
- ▶ New PA imposed that overlaps existing productive land and restricts land use
- ▶ **Assuming initially efficient allocation, what happens to net welfare?**
- ▶ **What if land rents increase outside the PA?**

Background: Conservation and Development

- ▶ On other hand, PAs may generate tourism income and other positive spillovers
- ▶ Surprisingly little evidence on income channel
- ▶ Main challenge: non-random placement of PAs

Sims (2010): Conservation and Development: Evidence from Thailand

- ▶ **Research Question:** How does PA policy affect local poverty outcomes?
 - ▶ What are the mechanisms?
- ▶ **Setting:** Thailand in 2000
- ▶ **Data:** Household consumption surveys + PA locations
- ▶ **Empirical Strategy:** Instrument PA location with priority watershed status
- ▶ **Results:** PAs increase consumption and lower poverty
 - ▶ Higher tourism is the main mechanism

- ▶ Wildlife sanctuaries (N=31) and national parks (N=57)
 - ▶ Source: World Database on Protected Areas
 - ▶ Includes year of establishment
- ▶ Poverty Mapping
 - ▶ Model hh income in 2000 Thai Economic Survey as a function of hh features + assets
 - ▶ Use coefficient to predict income of census households

- ▶ Poverty Measure

$$FGT(\alpha) = \frac{1}{n} \sum_{y_i < p} \left(\frac{p - y_i}{p} \right)^\alpha$$

where p is poverty line and y_i is income

Empirical Strategy: Ordinary Least Squares

$$y_{ij} = \beta_1 PA_{ij} + \beta_2 G_{ij} + \beta_3 X_{ij} + \alpha_j + \epsilon_j$$

- ▶ y_{ij} is socioeconomic outcome for locality i in district j
- ▶ PA_{ij} is share of locality land designated as PA prior to 2000
- ▶ G_{ij} are fixed geographic controls (slope, elevation, water, etc.)
- ▶ X_{ij} is a measure of pre-treatment (historic forest cover, distance to major roads)
- ▶ α_j are district fixed effects

What variation is exploited?

What are the identification assumptions?

What are potential omitted characteristics?

OLS Results

	Outcomes			
	(1)	(2)	(3)	(4)
	Log consumption (Baht/month)	Poverty headcount ratio	Poverty gap	Squared poverty gap
Share protected	0.135*** (0.033)	− 0.066*** (0.017)	− 0.017*** (0.006)	− 0.006** (0.002)
R^2	0.593	0.702	0.627	0.563
N localities	4113	4113	4113	4113

- ▶ Median land share protect is 1/3
- ▶ Expected $\uparrow$ in consumption is about 4.5% (divide by coeff. by 3)

Instrumental Variables Solution

- ▶ Instrument PA placement with **priority watershed status**
- ▶ Upper watersheds protected to improve **downstream** water quality
 - ▶ Watershed protection (PA) not driven by local (headwater) economic needs

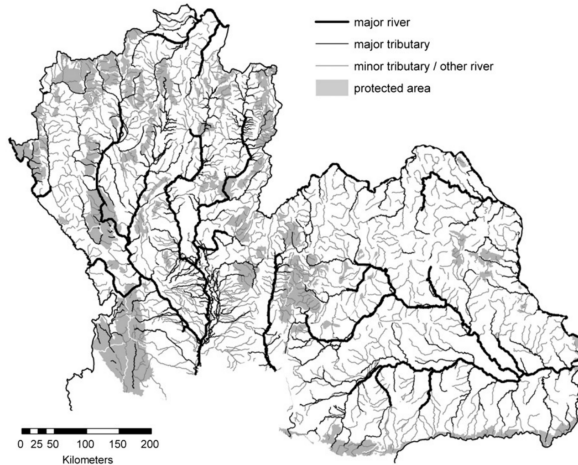
- ▶ First Stage:

$$PA_{ij} = \pi_1 Z_{ij} + \pi_2 G_{ij} + \pi_3 X_{ij} + \pi_4 W_{ij} + \delta_j + u_{ij}$$

- ▶ Z_{ij} includes a) i overlaps major tributary, b) distance to nearest major tributary .
- ▶ π_1 picks up variation in PA driven by differences in downstream characteristics

Watershed Instrument

a



Is the IV plausible? Why or why not?

IV Results

	Outcomes			
	(1) Log consumption (Baht/month)	(2) Poverty headcount ratio	(3) Poverty gap	(4) Squared poverty gap
IV estimates				
A. Intersects with tributary				
Share protected	0.372** (0.184)	−0.183** (0.092)	−0.063* (0.034)	−0.030* (0.017)
B. Distance to major tributary				
Share protected	0.358** (0.172)	−0.187** (0.093)	−0.069** (0.035)	−0.033* (0.017)
C. Both instruments				
Share protected	0.365** (0.170)	−0.185** (0.088)	−0.066** (0.033)	−0.031* (0.016)

- ▶ Signs agree with OLS, but magnitudes are more than 2x as large
- ▶ Consistent with idea that OLS negatively biased

Mechanisms: Is tourism responsible?

- ▶ PAs restricts land use, so why does income increase?
- ▶ Test 1: Test whether impacts in national parks $>$ in wildlife sanctuaries
 - ▶ Idea: NP's encourage tourism whereas sanctuaries only permit ecotourism/research
- ▶ Test 2: Back-of envelope calculation
 - ▶ if we know costs/benefits of tourism, is it enough to explain net $\uparrow$ in consumption?
- ▶ Test 3: Are benefits higher where tourism is higher
 - ▶ Park statistics show that visitation is highest near major cities

Test 1: National parks vs. Wildlife Sanctuaries

Socioeconomic outcomes and share protected by IUCN category.

	Outcomes			
	(1) Log consumption (Baht/mo)	(2) Poverty headcount ratio	(3) Poverty gap	(4) Squared poverty gap
Share WLS (I)	0.054 (0.050)	-0.019 (0.032)	-0.008 (0.013)	-0.004 (0.006)
Share NP (II)	0.086** (0.038)	-0.057*** (0.018)	-0.015*** (0.005)	-0.005** (0.002)
Share FR (IV)	-0.062*** (0.016)	0.022*** (0.008)	0.005* (0.003)	0.002 (0.001)
R^2	0.870	0.596	0.703	0.628
N localities	4113	4113	4113	4113

- Effects larger where tourism is allowed (NPs)
- Difference in coefficients not statistically significant

Test 2: Back of the envelope calculation

- ▶ Assume income is agricultural and reduced proportionally to land lost
- ▶ Assume credit constraints (no savings) $\rightarrow$ income losses = consumption losses
- ▶ Consumption cost of PAs is 11% of income (forest cover coeff. - see paper)
- ▶ To achieve 5% *net* $\uparrow$ in consumption, tourism must generate 16% consumption
 - ▶ Equivalent to 233 Baht per month ($0.16 \times$ mean consumption)
- ▶ 10.81 mil. tourists in 2020 and 172,978 hh in PA localities = 61 tourists/hh
- ▶ Each tourist would need to contribute: $(233 \times 12) / 61 = 46$ Baht (\$USD 1.20) of net income to generate observed increases in consumption

Is this reasonable?

- ▶ What do you think of the paper?
- ▶ Do you believe the instrument?
- ▶ What are some alternative explanations?